







# STATE TECHNICAL UNIVERSITY

END SEMESTER EXAMINATION 2025-2026 | ANSWER BOOKLET

CANDIDATE ROLL NO: 2026-CS-PH-007  
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SUBJECT & CODE: Physics (CS 201)  
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BARCODE SCAN TRACKING



\*BK-PH-80007\*

| Q.No  | Q1 | Q2 | Q3 | Q4 | Q5 |
|-------|----|----|----|----|----|
| Max   | 14 | 14 | 14 | 14 | 14 |
| Marks |    |    |    |    |    |

START WRITING YOUR ANSWERS BELOW. DO NOT WRITE IN THE MARGINS.

**Q.1** Ans 1. Group Velocity & Particle Velocity:

- A moving particle is represented by a de Broglie wave packet.

- Phase Velocity ( $v_p$ ): Speed of a single wave =  $\omega / k$ .

Since  $\omega = E/\hbar$  and  $k = p/\hbar$ ,  $v_p = E/p$ .

For non-relativistic particle,  $E = p^2 / 2m \Rightarrow v_p = p / 2m = v/2$ .

- Group Velocity ( $v_g$ ): Speed of the wave packet envelope:

$v_g = d(\omega) / dk = dE / dp$ .

$dE/dp = d(p^2 / 2m)/dp = p/m = v$  (particle velocity).

- Conclusion: Group velocity equals the particle velocity ( $v_g = v$ ).

The wave packet spreads over time due to dispersion, which is described by Schrodinger's equation.







**Q.2****Ans 2. Optical Interference & Alternative Strategies:**

1. Basic Principle: Superposition of coherent light beams producing bright and dark fringes.

2. Two Main Methods:

- Division of Wavefront: Incident wavefront is split spatially using two apertures.

Example: Young's Double Slit (YDSE). Fringe separation:  $\beta = \lambda \cdot D / d$ .

- Division of Amplitude: Splitting amplitude via partial reflection and transmission.

Example: Newton's Rings and Thin Films.

3. Thin Film Interference:

Optical path difference =  $2 \cdot \mu \cdot t \cdot \cos(r) \pm \lambda/2$ .

Condition for constructive interference (Bright fringe):  $2 \cdot \mu \cdot t \cdot \cos(r) = (2n - 1) \cdot \lambda/2$ .

Condition for destructive interference (Dark fringe):  $2 \cdot \mu \cdot t \cdot \cos(r) = n \cdot \lambda$ .

4. Newton's Rings: Fringes of equal thickness formed between plano-convex lens and glass plate.

Diameter of nth dark ring:  $D_n = \sqrt{4 \cdot n \cdot R \cdot \lambda}$ .







**Q.3****Ans 3. Lasers & Optical Fiber Technology:****A. Laser Fundamentals:**

- Three processes: Absorption, Spontaneous Emission, Stimulated Emission.
- Einstein's relation:  $B_{12} = B_{21}$  and  $A_{21}/B_{21} = 8\pi h \nu^3 / c^3$ .
- Population inversion:  $N_2 > N_1$  achieved using an optical pumping source and metastable state.
- Laser characteristics: High coherence, monochromaticity, directionality, and high brightness.

**B. Optical Fibres:**

- Structure: Cylindrical dielectric waveguide with Core ( $n_1$ ) and Cladding ( $n_2$ ),  $n_1 > n_2$ .
- Total Internal Reflection guides light along the core.
- Numerical Aperture:  $NA = \sqrt{n_1^2 - n_2^2} = \sin(\theta_{\max})$ .
- Applications: Telecommunication backbones, medical endoscopes, optical sensing.







**Q.4**      Ans 4. Electromagnetism and Maxwell's Field Equations:

## 1. Maxwell's Equations in Differential Form:

- 1.  $\text{div}(\mathbf{D}) = \rho$  (Gauss Law for Electric field)
- 2.  $\text{div}(\mathbf{B}) = 0$  (Gauss Law for Magnetic field)
- 3.  $\text{curl}(\mathbf{E}) = - \frac{d\mathbf{B}}{dt}$  (Faraday's Induction Law)
- 4.  $\text{curl}(\mathbf{H}) = \mathbf{J} + \frac{d\mathbf{D}}{dt}$  (Ampere's Law modified with Displacement Current  $\mathbf{J}_d$ )

2. Displacement Current:  $\mathbf{J}_d = \epsilon_0 \frac{d\mathbf{E}}{dt}$ . Maxwell added this term to maintain conservation of charge across capacitor charging cycles.

## 3. Boundary Conditions:

- $E_{1_t} = E_{2_t}$  (Tangential electric field continuous)
- $D_{1_n} - D_{2_n} = \rho_s$  (Normal displacement field)
- $B_{1_n} = B_{2_n}$  (Normal magnetic flux density continuous)
- $H_{1_t} - H_{2_t} = K$  (Tangential magnetic field)

4. Poynting Vector  $\mathbf{S} = \mathbf{E} \times \mathbf{H}$  gives the rate of energy transport by EM wave.







**Q.5**      Ans 5. Semiconductor Physics & Crystal Structure:

## 1. Crystal Lattices:

- SC (APF = 0.52), BCC (APF = 0.68), FCC (APF = 0.74).
- Miller Indices (hkl): Interplanar spacing  $d_{hkl} = a / \sqrt{h^2 + k^2 + l^2}$ .
- Bragg's Law:  $2 * d * \sin(\theta) = n * \lambda$ .

## 2. Band Theory:

- Periodic potential creates Valence Band, Conduction Band, and Energy Gap Eg.

## 3. Carrier Concentration:

- Total current density:  $J = q * (n * \mu_e + p * \mu_h) * E$ .
- Fermi-Dirac distribution:  $f(E) = 1 / [ 1 + \exp((E - E_f)/kT) ]$ .

## 4. Direct vs Indirect Bandgap:

- Direct bandgap semiconductors (GaAs, InP) emit light efficiently upon electron-hole recombination.
- Indirect bandgap semiconductors (Si, Ge) involve lattice vibrations (phonons).

--- End of Answer Booklet (All 5 questions attempted) ---

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